ChatGPT

Multi-region latent prediction as low-rank operator regression

Keywords and abstract

0) **Keywords.** Latent prediction; operator regression; rank constraints; self-supervised theory; structured occlusions; identifiability; multitask prediction.

0) **Abstract.**
Joint-Embedding Predictive Architecture (JEPA) predicts a target embedding from a context embedding. [1]
We study within-sample latent prediction when several disjoint regions are withheld. To prove.
We frame the population objective as estimating a linear operator from context embeddings to target embeddings. To prove.
We show the operator effectively factors through an embedding dimension d, imposing a rank bottleneck. To prove.
We derive conditions where predicting $K!\ge!2$ target regions increases the identifiable predictive subspace. To prove.
We state a finite-sample rate using singular-subspace perturbation bounds under sample noise. To prove.
We predict downstream accuracy improves with K until a saturation threshold near an effective rank. To prove.
We predict overlap between target regions reduces the marginal gain from increasing K. To prove.
We validate the consequences on a linear-Gaussian synthetic generator with controllable heterogeneity. To prove.

Setup and objective

1) **Definitions in required order.**
Joint-Embedding Predictive Architecture (JEPA): predict a target embedding from a context embedding. [1]
Mask: an index set selecting observed coordinates from one sample. To prove.
Mask distribution: a probability distribution over masks. To prove.
Block mask: a mask supported on structured contiguous index sets. To prove.
Block-masked JEPA (BM-JEPA): a JEPA where context and targets are defined by masks. To prove.
Multi-target BM-JEPA: BM-JEPA with $K\ge 2$ disjoint or overlapping target blocks. [2]
Context encoder: f_θ mapping context content to a context embedding. [3]
Target encoder: $f_{\bar{\theta}}$ mapping target content to a target embedding. [3]
Predictor: g_ϕ mapping a context embedding and target metadata to a predicted target embedding. [2]
Stop-gradient: an operator that blocks gradients through teacher targets. [4]
EMA teacher: $\bar{\theta}$ updated as an exponential moving average of θ. [3]

Reduced-rank regression (RRR): multivariate regression with rank-constrained coefficients. [5]
Cross-covariance operator: Σ_{YH} between targets and context embeddings. To prove.

1) **Data model and coordinate system.**
Let one sample be $X\in\mathcal{X}$ with coordinate index set $\Omega={1,\dots,p}$. To prove.
A mask $m\subseteq\Omega$ reveals the restricted content X_m. To prove.
We permit non-vector data by assuming X_m is a measurable object for each m. To prove.

1) **Mask model and randomness sources.**
A mask tuple is $(m_c,m_{t,1},\dots,m_{t,K})\sim\mathcal{D}$ for a mask distribution $\mathcal{D}$. To prove.
Randomized objects are: $X\sim\mathbb{P}$, masks from $\mathcal{D}$, minibatching, and SGD noise. To prove.
Fixed objects are: embedding dimension d, target count K, hypothesis classes for (f_θ,g_ϕ). To prove.

1) **BM-JEPA objective under squared loss with stop-gradient.**
For each $k\in{1,\dots,K}$ define

$$H := f_\theta(X_{m_c}) \in \mathbb{R}^d, \qquad Y_k := f_{\bar\theta}(X_{m_{t,k}}) \in \mathbb{R}^d.$$

To prove.

Let target metadata be $z_k:=\operatorname{meta}(m_{t,k})$. To prove.
Define the prediction $\widehat Y_k := g_\phi(H,z_k)$. To prove.

\begin{align} \mathcal{L}*{\text{BM-JEPA}}(\theta,\phi;\bar\theta) := \mathbb{E}* \Bigg[\frac{1}{K}\sum_{k=1}^K \big|\widehat Y_k-\operatorname{sg}(Y_k)\big|_2^2\Bigg]. \end{align} To prove.},\, (m_c,m_{t,1:K}) \sim\mathcal{D}

Squared loss matches the image instantiation’s latent regression form with an $L2$ distance. [2]
Stop-gradient and an EMA target encoder are used in a video instantiation to prevent collapse. [4]

1) **Frozen teacher specialization.**
Frozen teacher means $\bar\theta$ is fixed during optimization of (θ,ϕ). To prove.
This matches the teacher–student separation used when targets come from a distinct, slowly updated branch. [6]

1) **EMA teacher extension.**
EMA teacher updates $\bar\theta\leftarrow \tau\bar\theta+(1-\tau)\theta$ after each student update. [3]
The video instantiation explicitly uses an EMA encoder on the target branch. [4]

Main theorem set

2) **Notation for theorems.**
Let $Y:=(Y_1^\top,\dots,Y_K^\top)^\top\in\mathbb{R}^{Kd}$. To prove.

Let $\Sigma_{HH}:=\mathbb{E}[HH^\top]$ and $\Sigma_{YH}:=\mathbb{E}[YH^\top]$. To prove.
Define the **predictive rank** as $r_\star:=\operatorname{rank}(\Sigma_{YH})$. To prove.
Define the **coverage matrix** $C:=\mathbb{E}[\operatorname{diag}(\mathbf{1}_{m_c})]\in\mathbb{R}^{p\times p}$. To prove.

2) **Theorem A: Reduced-rank characterization of the population minimizer.**
Assume f_θ is linear in X_{m_c} and g_ϕ is linear in H. To prove.
Assume the predictor's coefficient matrix is constrained to rank at most $r\le d$. To prove.
Then the population minimizer is the reduced-rank regression operator obtained by truncating the unconstrained regression map. To prove.
The truncation keeps the top-r canonical modes between H and Y. To prove.
This matters because d fixes an upper bound on achievable predictive rank. To prove.

2) **Theorem B: Multi-target identifiability improvement from heterogeneity.**
Assume the frozen teacher and fixed $(f_\theta,f_{\bar\theta})$ induce $(H,Y_1,\dots,Y_K)$. To prove.
Assume a heterogeneity condition: the left singular subspaces of Σ_{Y_kH} span dimension $r_\star$ with minimal singular value $\sigma_{\min}(\operatorname{stack}k \Sigma>0$. To prove.})\ge \lambda_{\text{het}}
Then increasing K can increase $r_\star$ up to d, improving identifiability of the predictive subspace. To prove.
The improvement scales with λ_{het} and degrades with target overlap. To prove.
This matters because multi-region targets can remove degeneracies present with $K=1$. To prove.

2) **Theorem C: Capacity lower bound from effective rank limitations.**
Assume $r_\star>d$ and the student representation has dimension d. To prove.
Then any BM-JEPA solution has population risk at least the spectral tail beyond the top-d predictive directions. To prove.
Equivalently, the best achievable error lower-bounds a function of ${\sigma_{j}(\Sigma_{YH})}_{j>d}$. To prove.
This matters because it mechanizes "embedding dimension limits achievable predictive accuracy." To prove.

2) **Theorem D: Falsifiable saturation prediction in K.**
Assume Theorem B's heterogeneity and Theorem C's capacity bound hold. To prove.
Then performance improves with K until $\min(Kd,r_\star)\approx d$ is reached. To prove.
Beyond that point, additional targets yield diminishing returns under fixed d. To prove.
This matters because it predicts a measurable saturation curve in K. To prove.

2) **Supporting lemmas.**
Lemma A: Under squared loss, the optimal unconstrained linear predictor is $\Sigma_{YH}\Sigma_{HH}^{-1} H$. To prove.
Lemma B: Under rank $\le r$, the best linear predictor equals the best rank-r approximation in a whitened space. To prove.
Lemma C: Empirical $\widehat \Sigma_{YH}$ concentrates in operator norm at rate $\tilde O(\sqrt{\operatorname{effdim}/n})$. To prove.
Lemma D: Principal angles between empirical and population singular subspaces are controlled by Wedin-type bounds. [7]

Lemma E: If target blocks overlap heavily, stacked cross-covariance becomes low-rank even for large K. To prove.

Claims-to-evidence map

3) **Mini table.**

Claim	Evidence	Dependencies	Failure modes
BM-JEPA is embedding regression with a low-rank bottleneck.	Theorem A.	To prove.	Nonlinear g_ϕ can evade a linear rank bottleneck.
Multi-region targets increase identifiable predictive rank under heterogeneity.	Theorem B.	To prove.	Targets become redundant under strong overlap or symmetry.
Finite-sample subspace error is controlled by singular-subspace perturbation bounds.	Lemma D. [7]	Concentration for $\widehat \Sigma_{YH}$ is needed. To prove.	Heavy-tailed embeddings can violate operator-norm concentration.
Too-small d forces irreducible error set by predictive spectrum tails.	Theorem C.	To prove.	If f_θ adapts, $r_\star$ itself may shrink.
Accuracy saturates in K beyond an effective-rank threshold.	Theorem D.	To prove.	If increasing K changes learned features, saturation can shift.
Multi-region masking improves semantic quality in practice.	Empirical ablation in the image instantiation. [8]	Unclear for other modalities without matching ablations.	In low-context regimes, many targets may induce shortcut learning.

Proof plan

4) **Proof dependency order.**
We prove Lemma A, then Lemma B, then Lemma C. To prove.
We then invoke Lemma D for subspace perturbation control. [7]
We conclude Theorem A, then Theorem B, then Theorem C, then Theorem D. To prove.

4) **Lemma A.**
Statement. The optimal unconstrained linear predictor is $\widehat Y=\Sigma_{YH}\Sigma_{HH}^{-1}H$. To prove.
Why needed. It identifies the baseline operator before rank constraints. To prove.

Route. Differentiate population risk, set gradient to zero, solve normal equations. To prove.
Obstacle. Handle singular Σ_{HH} via pseudoinverse and stability conditions. To prove.

4) **Lemma B.**
Statement. Rank-r constrained minimizer equals top-r canonical modes after whitening. To prove.
Why needed. It connects BM-JEPA to RRR and CCA algebra. To prove.
Route. Reduce to RRR by whitening H, then apply reduced-rank approximation. To prove.
Obstacle. Track dependence on K when stacking multi-target outputs. To prove.

4) **RRR / CCA connections and exact algebraic conditions.**
RRR is classically linked to canonical variates in multivariate regression settings. [5]
CCA can be expressed through block covariance matrices and generalized eigenproblems. [9]
We must pin down the exact whitening operators matching the BM-JEPA stacked output. To prove.

4) **Lemma C.**
Statement. $|\widehat\Sigma_{YH}-\Sigma_{YH}|_{\mathrm{op}}$ concentrates with effective dimension dependence. To prove.
Why needed. It supplies finite-sample error entering a perturbation bound denominator. To prove.
Route. Use matrix Bernstein on outer products $YH^\top$ after truncation. To prove.
Obstacle. Embeddings may be non-subgaussian under learned encoders. To prove.

4) **Matrix perturbation for singular subspaces.**
Wedin's theorem bounds principal angles for singular subspaces under perturbations. [7]
A Davis–Kahan variant yields eigen-subspace bounds with population eigengap denominators. [10]
We must choose the rectangular form matching Σ_{YH}, not symmetric covariance only. To prove.

4) **Lemma D.**
Statement. Subspace distance obeys $|\sin\Theta(\widehat U,U)|\lesssim |\widehat\Sigma_{YH}-\Sigma_{YH}|/\text{gap}$. [7]
Why needed. It turns covariance estimation into identifiable rank claims. To prove.
Route. Apply Wedin to a rank-r truncation of Σ_{YH}. [7]
Obstacle. Define a population singular gap that is checkable from data. To prove.

4) **Mask-coverage condition that is non-vacuous and checkable.**
Empirically, low coverage makes prediction trivial and harms downstream performance in video feature prediction. [11]
We will formalize coverage using $C=\mathbb{E}[\operatorname{diag}(\mathbf{1}{m_c})]$ and require $ \lambda(C)\ge c_0>0$. To prove.
We will define a target-coverage analogue for each $m_{t,k}$ and aggregate across k. To prove.
Obstacle: block masks induce structured dependence across coordinates. To prove.

4) **Heterogeneity condition on targets that is minimal and interpretable.**
Masked prediction identifiability can depend on how many tokens are predicted. [12]
We will define heterogeneity as a lower bound on the smallest nonzero singular value of the stacked cross-covariance. To prove.
We will show this condition is satisfied when target blocks probe distinct latent factors. To prove.
Obstacle: target encoders can collapse heterogeneity by construction. To prove.

4) **Theorem B proof sketch.**
Step 1. Write Σ_{YH} as a stack of Σ_{Y_kH}. To prove.
Step 2. Use heterogeneity to lower-bound $\sigma_r(\Sigma_{YH})$ as K increases. To prove.
Step 3. Apply Lemma D to show stable recovery of the predictive subspace. [7]
Main obstacle. Quantify how overlap in target blocks reduces λ_{het}. To prove.

4) **Theorem C proof sketch.**
Step 1. Express risk as squared error under best rank-d operator approximation. To prove.
Step 2. Lower-bound error by spectral tail beyond dimension d. To prove.
Main obstacle. Separate approximation error from estimation error under learned f_θ. To prove.

Filler theory test

5) **Design constraints implied by the theory.**
If d is too small, the capacity bound predicts irreducible error. To prove.
Therefore, increasing K cannot substitute for increasing d beyond saturation. To prove.

5) **Ablation prediction one.**
Increasing K improves performance until $\min(Kd,r_\star)\approx d$, then saturates. To prove.
This prediction is falsifiable by varying K while keeping architecture and masks fixed. To prove.
The image instantiation already computes predictions for multiple target blocks per sample. [2]

5) **Ablation prediction two.**
For fixed K, increasing overlap among target blocks decreases performance gains. To prove.
This prediction follows because overlap reduces the stacked cross-covariance's effective rank. To prove.
The image instantiation explicitly allows possibly overlapping target blocks, enabling this ablation. [2]

5) **Ablation prediction three.**
Mask coverage must remain high enough to avoid trivial prediction tasks. [11]
Yet coverage must also leave enough context information for nondegenerate prediction. To prove.
Video feature prediction reports best results under multi-block masking and high coverage settings. [13]

5) **Ablation prediction four.**
If the EMA rate τ is too large, targets change too slowly and reduce learning signal. To prove.
If τ is too small, the teacher tracks student too closely and risks collapse. To prove.
Non-contrastive theory attributes collapse avoidance to stop-gradient plus EMA interactions. [14]

Related work and novelty positioning

6) **Closest prior work by component.**
Generic JEPA is defined as predicting a representation of y from a representation of x. [1]
The image instantiation predicts multiple target regions from a single context region using $L2$ loss. [2]
The video instantiation uses stop-gradient and an EMA encoder, and uses multi-block sampling. [4]
A multimodal latent-prediction framework predicts teacher representations from a masked input with an EMA teacher. [15]
Collapse-prevention mechanisms for predictor plus stop-gradient and EMA are analyzed in linear networks.

[16]

Deep linear self-distillation compares JEPA objectives to reconstruction objectives in learning dynamics. [17]
Masked prediction can yield parameter identifiability, and predicting more tokens affects difficulty. [12]
Reduced-rank regression relates reduced-rank multivariate regression to canonical variates. [5]
Singular-subspace perturbation control is available via Wedin's theorem and refinements. [7]

6) **Eight nearest papers with audit-ready metadata.**

- **Yann LeCun [18] (OpenReview, 2022): "A Path Towards Autonomous Machine Intelligence." [1]
  Covers generic JEPA definition and motivation for prediction in representation space. [1]
  Does not cover multi-region intra-sample objectives or statistical rates. To prove.

- **Mahmoud Assran [19] et al. (CVPR, 2023): image instantiation with multi-block targets and $L2$ regression. [2]
  Covers multi-target prediction and ablations showing multi-block helps semantics. [8]
  Does not cover reduced-rank operator interpretations or identifiability conditions. To prove.

- **Adrien Bardes [20] et al. (TMLR, accepted 2024): video instantiation with stop-gradient, EMA, and multi-block masking ablations. [21]
  Covers explicit use of stop-gradient and EMA as collapse prevention within JEPA training. [4]
  Does not cover rank-based identifiability improvements from multiple target regions. To prove.

- **Alexei Baevski [22] et al. (ICML, 2022): predict teacher representations from masked inputs using EMA teacher. [15]
  Covers masked latent prediction across modalities with teacher–student regression. [15]
  Does not isolate multi-region heterogeneity as rank-identification driver. To prove.

- **Yuandong Tian [23] et al. (ICML, 2021): theory for non-contrastive self-distillation with stop-gradient and EMA. [16]
  Covers collapse avoidance roles of predictor, stop-gradient, and EMA in linear models. [14]
  Does not cover structured intra-sample multi-region objectives. To prove.

- **Etai Littwin [24] et al. (NeurIPS, 2024): deep linear analysis comparing JEPA and MAE dynamics. [17]
  Covers a JEPA-versus-reconstruction theory with explicit dynamical theorems. [17]
  Does not cover multi-target block heterogeneity or operator-rank estimation. To prove.

- **Bingbin Liu [25] et al. (NeurIPS, 2022): masked prediction identifiability depends on task design and token count. [12]
  Covers identifiability as a function of how many tokens are predicted. [12]
  Does not cover embedding-to-embedding JEPA objectives or EMA teachers. Unclear.

- **Zijian Dong [26] et al. (NeurIPS, 2024): JEPA on fMRI with tailored spatiotemporal masking. [27]
  Covers JEPA instantiation beyond images, using masking and EMA with a predictor. [28]
  Does not cover regression-rank or identifiability theory. To prove.

6) **Strongest JEPA-versus-reconstruction theory, and what it misses.**
A deep linear theory explicitly compares JEPA and MAE training dynamics and critical times. [17]
It models self-distillation JEPA, not structured multi-region masking within one sample. [29]
It explains feature prioritization, not multi-target identifiability from heterogeneous blocks. To prove.

6) **Strongest deep linear self-distillation theory, and what it misses.**
The deep linear work provides dynamical theorems for JEPA training in linear networks. [17]
It does not derive finite-sample operator estimation rates under block-structured masking. To prove.

6) **“A+B disguised as novelty” check.**
Let **A** be classical RRR/CCA plus subspace perturbation tools. [5]
Let **B** be BM-JEPA multi-region latent regression used in modern instantiations. [2]
The genuinely new part is a minimal heterogeneity condition linking multi-target masking to predictive rank growth. To prove.
The second new part is a falsifiable saturation law coupling K, d, and predictive spectrum tails. To prove.
If heterogeneity can be reduced to standard multi-task low-rank MTL conditions, novelty risks collapse. Unclear.

6) **Missed-citation risks.**
The IJCV version of Context Autoencoder is paywalled here, so exact claims beyond arXiv are Unclear. [30]
The accepted TMLR metadata is visible, but the downloadable PDF still shows an anonymous header here. [21]
Search queries for closing gaps: “TMLR final PDF Revisiting Feature Prediction OpenReview pdf camera-ready,” and “IJCV s11263-023-01852-4 PDF open access.” Unclear.

Minimal experiments

7) **Experiment one: K-saturation under a fixed bottleneck.**
Generate latent state $S\sim\mathcal{N}(0,I_{r_\star})$ and blocks $X_{b}=A_b S+\varepsilon_b$. To prove.
Define the teacher target embedding as $Y_k=B_k X_{b_k}$ and context embedding as $H=B_c X_{b_c}$. To prove.
Train the optimal linear BM-JEPA predictor with embedding dimension d and vary K. To prove.
Measure risk and subspace error versus K, expecting concave gains then saturation. To prove.

7) **Experiment two: heterogeneity-controlled identifiability.**
Fix K and vary principal angles between $\operatorname{col}(A_{b_k})$ across targets. To prove.
Keep coverage constant while changing heterogeneity, keeping $r_\star$ fixed in the generator. To prove.
Measure recovered subspace angle to the true predictive subspace, expecting monotone improvement with heterogeneity. To prove.

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